

Systems of Linear Equations



Setting up the augmented matrix

- 1 Write the augmented matrix for the system $2x + y - z = 3$, $x - 3y + 2z = -1$, $4x + y + z = 5$.
-

Elementary row operations

- 2 List the three types of elementary row operations.
-

Solving 2x2 systems by elimination

- 3 Solve $2x + 3y = 7$, $x - y = 1$.
- 4 Solve $3x - 2y = 4$, $5x + y = 11$.
-

Solving 3x3 systems by Gaussian elimination

- 5 Solve $x + y + z = 6$, $2y + 5z = -4$, $2x + 5y - z = 27$.
-

Why row operations preserve the solution set

- 6 Explain briefly why multiplying a row by a nonzero constant c doesn't change a system's solution set.
- 7 Why is the restriction $c \neq 0$ needed for this row operation?
-

More 2x2 elimination practice

- 8 Solve $4x + y = 10$, $2x - y = 2$.
- 9 Solve $x + 2y = 5$, $3x - y = 1$.
-

More 3x3 Gaussian elimination practice

- 10 Solve $2x + y + z = 5$, $x - y + 2z = 1$, $3x + 2y - z = 8$.
-

Systems with no solution or infinitely many solutions

- 11 Solve $x + y = 3$, $2x + 2y = 8$, and explain what you find.
- 12 Solve $x - y = 1$, $2x - 2y = 2$, and explain what you find.

Setting up a system from a word problem

- 13 A store sells adult tickets for 8 dollars and child tickets for 5 dollars. On a day when 90 tickets were sold for 600 dollars total, set up a system of equations for the number of adult (a) and child (c) tickets sold.
-

Solving the ticket-sales system

- 14 Solve the system from the previous question.
-

Reading a solution from row echelon form

- 15 The augmented matrix $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 5 \end{array} \right]$ is already in reduced row echelon form. State the solution.
-

Verifying a solution

- 16 Verify that $(x, y, z) = (1, 2, -1)$ solves $x + y + z = 2$, $2x - y + z = -1$, $x + 2y - 3z = 8$.
-

A system with a parameter

- 17 For which value of k does $x + 2y = 3$, $3x + ky = 9$ have infinitely many solutions?
-

Consistency of a system

- 18 A system of two linear equations in x, y is consistent if it has at least one solution. Geometrically, what does an inconsistent system of two lines look like?
- 19 Is the system $2x - y = 3$, $4x - 2y = 7$ consistent?
-

Number of solutions from row echelon form

- 20 A row echelon form has a row $\left[\begin{array}{ccc|c} 0 & 0 & 0 & 5 \end{array} \right]$. What does this tell you about the system?
- 21 A 3×3 system row-reduces to a row echelon form with a row of all zeros (no contradiction) and only 2 nonzero pivot rows. What does this tell you?
-

Back-substitution from an upper-triangular system

- 22 Solve by back-substitution: $x + 2y - z = 4$, $y + 3z = 5$, $2z = 6$.
-

Homogeneous systems

- 23** A system is homogeneous if every equation equals 0 (e.g. $2x - y = 0$, $x + 3y = 0$). Explain why $x = y = 0$ is always a solution of a homogeneous system.
- 24** Solve the homogeneous system $2x - y = 0$, $x + 3y = 0$.